

Audio/Video Processors for Embedded Multimedia Designs

Diversity of audio/video content and formats has burdened multimedia designers with the responsibility to adapt and deliver to consumer expectations. Audio/video processors are at the heart of these multimedia designs, and this story looks at what has changed in the last year

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Over the last few years, we have seen a revolution in the field of entertainment devices. The list that began with traditional radio and TV sets has exploded into an endless number of devices ranging from smartphones, digital cameras and camcorders, portable media players, mobile Internet devices (MIDs), notebooks, all the way up to large flat-panel displays, home theatre systems and much more.

Now, for most entertainment devices, processing audio and video is one of the most important functions, and selecting an optimal processing solution is one of the keys to success for any product that has to do any significant amount of computing. The desired product features influence the product cost, power consumption and performance as well.

Aspects to be considered

Audio processors take comparatively lower processing power than video processors, but they are both equally important. Let us take a look.

Audio processors. Audio processors serve a variety of fields, and each of these fields has its own challenges and design goals. In some fields, digital signal processing is used to produce high-fidelity sounds such as in the

entertainment industry where audio quality is paramount. On the other hand, communication systems require the audio to be clear while keeping within a low data rate. While designing any audio processing system, the

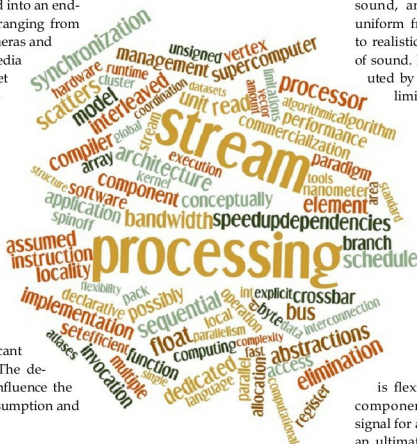
suppress external as well as electrical noise produced within the system itself, controlling the reverberation of acoustics produces good intelligibility.

Fidelity. Fidelity of sound is the overall frequency response of the sound, and a wide and relatively uniform frequency range contributes to realistic and precise augmentation of sound. Fidelity is basically contributed by every component, and any limitation at any point can affect the fidelity of the entire system.

Video processors. Video processing applications are growing exponentially, with the new kind of video-centric products surfacing rapidly. Computationally demanding video processing has different requirements for different applications. For instance, video applications such as home theatre systems require a processor which

is flexible enough to connect all components together, process the signal for a large living room, creating an ultimate home theatre experience by delivering premium sound quality.

On the other hand, vehicle-reversing cameras and other small-screen LCD applications for both automotive and non-automotive electronics require good image clarity. These applications also pose challenges such as safety and reduced power consumption, especially in automotive systems in order to minimise the power burden on the battery. This increasing range of



designers have three primary targets to achieve: good audibility, intelligibility and fidelity.

Audibility. The audibility of speech or music must be sufficient to achieve the desired effect attained without distortion or feedback.

Intelligibility. Intelligibility is determined by the signal-to-noise ratio and direct-to-reverberant ratio at the listener's end. Whilst the system must